

and uses non-toxic fuels. It has high specific energy and a power level of 1.2 megawatts demonstrated in Airborne Laser (ABL) programme. It operates at shorter lasing wavelength of 1.3 microns that has good atmospheric transmission. It requires a smaller aperture laser beam director and offers good target coupling efficiency. Low-pressure operation makes it better suited to airborne applications. Complex pressure recovery system is required for ground based applications.

ABL programme of the US Air Force uses a high-power chemical oxy-iodine laser system. ABL is one part of a layered ballistic missile defence system that addresses the ever-



Fig. 8: ABL concept

increasing ballistic missile threat. It is designed to destroy the hostile missile while it is still in the vulnerable boost phase of its flight (Fig. 8).

ABL uses an onboard surveillance system to detect and track the missile after launch. The beam-control system locks on to the target and then fires the high-energy laser. The entire system, that is, the high-energy laser, surveillance system and beam-control/fire-control system, is configured on a Boeing 747-400 fighter aircraft.

ABL made its maiden flight in July 2002. In January 2010, the high-energy laser was used in-flight, to intercept, although not destroy, a test missile in the boost phase of flight. In February 2010, the system successfully destroyed a liquid-fuel-boosting ballistic missile. In December 2011, ABL programme was reportedly shelved as it was considered to be operationally not viable.

AGIL overcomes the disadvantage of heavy, aqueous based chemistry of a chemical oxy-iodine laser device and the undesirable wavelength range of HF laser that is strongly absorbed by the atmosphere. Once scaled to higher power levels, it would be a potent laser technology for the Space-borne laser programme.

Free electron lasers

Free electron laser is a unique laser, where the process of light amplification is a bit unconventional *vis-à-vis* what we have seen in respect of hitherto discussed lasers. Unlike conventional lasers that rely on bound atomic or molecular states, free electron lasers use a relativistic electron beam as the active medium. In the case of a free electron laser, lasing medium is a beam of free electrons completely unattached to any atoms.

The free electron laser is tunable over a wide range of wavelengths, from X-rays to microwaves. Tunability is achieved by varying electron energy. The accelerated beam of electrons having desired energy can be obtained from well-established technology of charged particle accelerators.

Free electron lasers, though thought of as potential sources for DEW applications, have their problems when it comes to generating higher powers at shorter wavelengths. Smaller free electron lasers with power output up to 100 watts are particularly important for research and medical applications.

Solid-state lasers

Solid-state lasers are electrically-driven devices using a solid-state laser gain medium. Gain medium is pumped by either flash lamps or laser diode bar arrays, with the latter being far more widely used in the state-of-the-art solid-state lasers.

The all-solid-state configuration offers unmatched advantages in



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